

Monthly Intelligence Report

Edition 01 — Cartago & the Pacific Ring of Fire Eruption Cycle

Where Costa Rica operates an SIEPAC–interconnected grid with zero land interconnects, ~99% renewable (89% in 2024 El Niño) generation (70.5% hydro + 29.4% geothermal), and the Guanacaste province has been in an active eruption cycle since 19 Mar 2021 — the 2023–2024 Turrialba 2014–2019 ash eruption damaged the Miravalles geothermal plant infrastructure. Inaugural edition · August 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

687

ICE TSO + single–DSO–dominant panel (ICE + CNFL + ESPH + JASEC + 4 cooperatives)

GRID LINES MAPPED

~1,428

230 / 138 / 34.5 kV + MV (SIEPAC ring)

MC SIMULATIONS

35.02 M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation–level engineering metrics with socio–economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). For Costa Rica, its engine processes 74,000+ source data points per run, executes 35.02 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 830+ million arithmetic operations — producing 210,000+ output data points with full uncertainty quantification across 169 substations and ~4,261 grid lines.

Operated as a non–profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30+ verified public sources — ICE network statement, ARESEP +

quality reports, CNFL + ICE distribution + Coopelesca/Coopeguanacaste cooperative filings, INEC regional accounts, BCCR national accounts, IMN consolidated meteorology + hydrology + OVSICORI seismic + volcanic monitoring, MINAE air quality, CSIRT-CR + MICITT (ECOI) cyber posture — making the methodology fully auditable and reproducible. Costa Rica is the inaugural single-country onboarding post-cohort-closure (Session 33B), deployed on the 13 August 2026 anchor as the first OECD founding-member country onboarded since the CEE-South-2026 architectural-acceptance run.

Substation in Focus

ICE – Central Telefónica

Heredia, heredia · 230 kV

0.586

High

0.110

86th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE

COMPONENTS	MODIFIERS
C Continuity: 0.462 / 0.30 (154%)	R3 Consequence: 1.061 ↑ amplifies
I Infrastructure: 0.278 / 0.25 (111%)	R4 Graph Criticality: 1.000 → neutral
S Saturation: 0.516 / 0.20 (258%)	R6a Restoration: 0.060 ↓ dampens
V Voltage: 0.614 / 0.10 (614%)	R6b Seismic: 0.160 ↓ dampens
E Economic: 0.586 / 0.10 (586%)	R7 Digital Readiness: 1.025 ↑ amplifies
T Transition: 0.098 / 0.05 (195%)	

This month's spotlight — automatically selected for June 2026 — is **ICE – Central Telefónica** in Heredia, heredia. With an R_final of 0.586 (High band, 86th fleet percentile), its risk profile is dominated by the V (Voltage) component at 614% of its weight ceiling, followed by E (Economic) at 586%. Modifiers collectively shift R_base by 31%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Costa Rica's Ley 8454 (2005) + National Cybersecurity Strategy 2023 (transposed via the EEA process, in force 1 January 2026) and MICITT (ECOI) competent-authority designation require, with CSIRT-CR as the operational CSIRT.

CYBER HIGH EXPOSURE

0

BLIND SPOTS

22

AVG R7 MODIFIER

1.0250

0.0% of fleet

High/Critical + R7 < median

Range [0.99, 1.04]

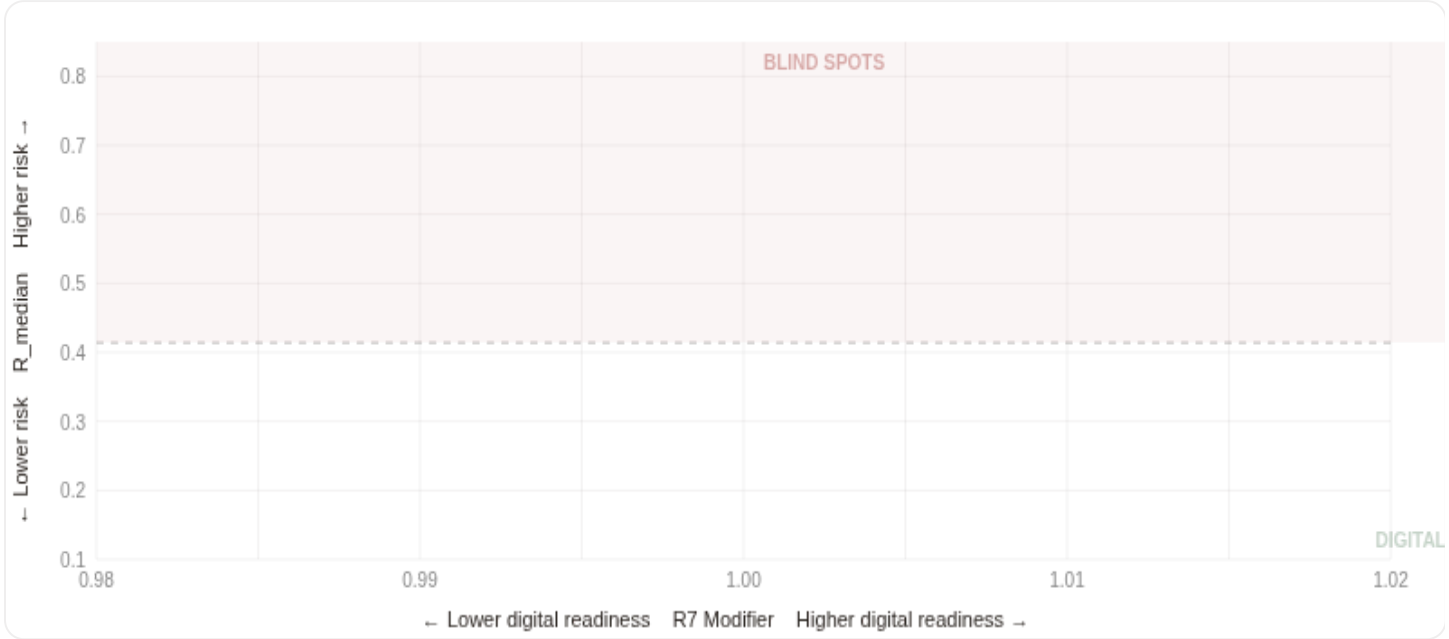
HIGH-CONSEQUENCE SUBS

43

25.4% · R3 ≥ 1.05

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **22 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 1.0250$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **alajuela (5), san-jose (5), guanacaste (5)**. Across the full fleet, 0 (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

Capital-Intensive / High-Consequence

Est. VoLL: ₡7,000–14,000/kWh (CRC)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

43 substations (25.4%)

High/Critical: **12** (27.9%)

Avg R: **0.423**

Avg E: **0.423** / 0.10

Low 0

Med 31

High 12

Crit 0



Industrial / Medium–Large Enterprise

Est. VoLL: €3,500–7,000/kWh (CRC)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

45 substations (26.6%) High/Critical: 15 (33.3%) Avg R: 0.452 Avg E: 0.452 / 0.10



Commercial / SME–Dense

Est. VoLL: €1,400–3,500/kWh (CRC)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

38 substations (22.5%) High/Critical: 10 (26.3%) Avg R: 0.401 Avg E: 0.401 / 0.10



Light / Agricultural / Rural

Est. VoLL: €500–1,400/kWh (CRC)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

43 substations (25.4%) High/Critical: 14 (32.6%) Avg R: 0.420 Avg E: 0.420 / 0.10



Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **12 substations** combine capital-intensive economic fabric ($R3 \geq 1.05$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **alajuela (19), san-jose (12), heredia (6)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 7.3%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 Provincia Digital Readiness Ranking

The 7 provincias ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI OECD–Americas sub-dimensions + CSIRT–CR maturity). Cross-referenced with average R_{median} to identify provincia combining digital exposure with high grid stress; Limón + Puntarenas carry the largest fibre/density penalty relative to San José.



WORST DIGITAL READINESS — TOP 15 PROVINCIA REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Heredia ▲ high R		1.0249
2	Alajuela ▲ high R		1.0249
3	San José		1.0250
4	Guanacaste ▲ high R		1.0250
5	Puntarenas		1.0251
6	Limón		1.0251
7	Cartago ▲ high R		1.0251

Provincia-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Heredia** has the lowest average R7 (1.0249), while **Cartago** leads at 1.0251 — a spread of 0.0002 across the R7 range. **2 provincia regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 1.0250, median = 1.0250; 0 substations classified HIGH cyber-exposure; 22 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **12 verified public data sources** feeding 95 variables across 6 components and 8 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

12

95 variables

WEEKLY / MONTHLY

2

High-frequency feeds

QUARTERLY / ANNUAL

8

Regular refresh cycle

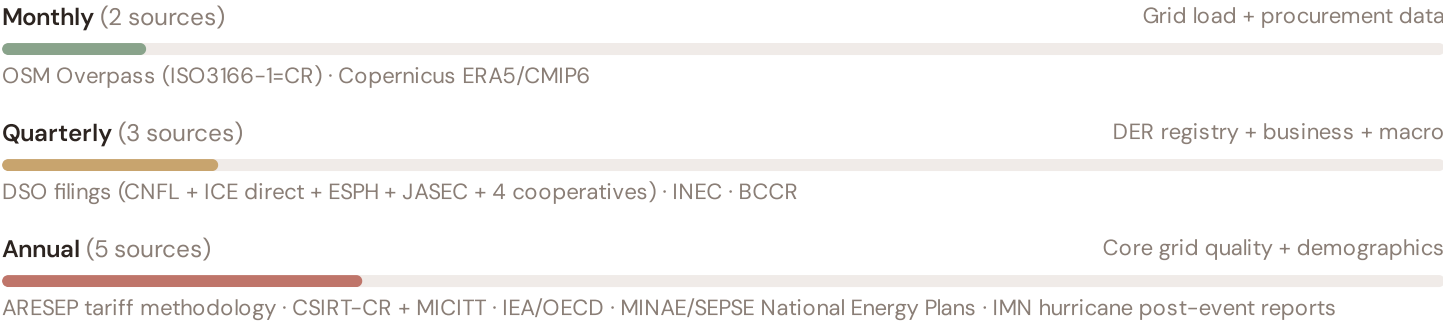
STATIC / DERIVED

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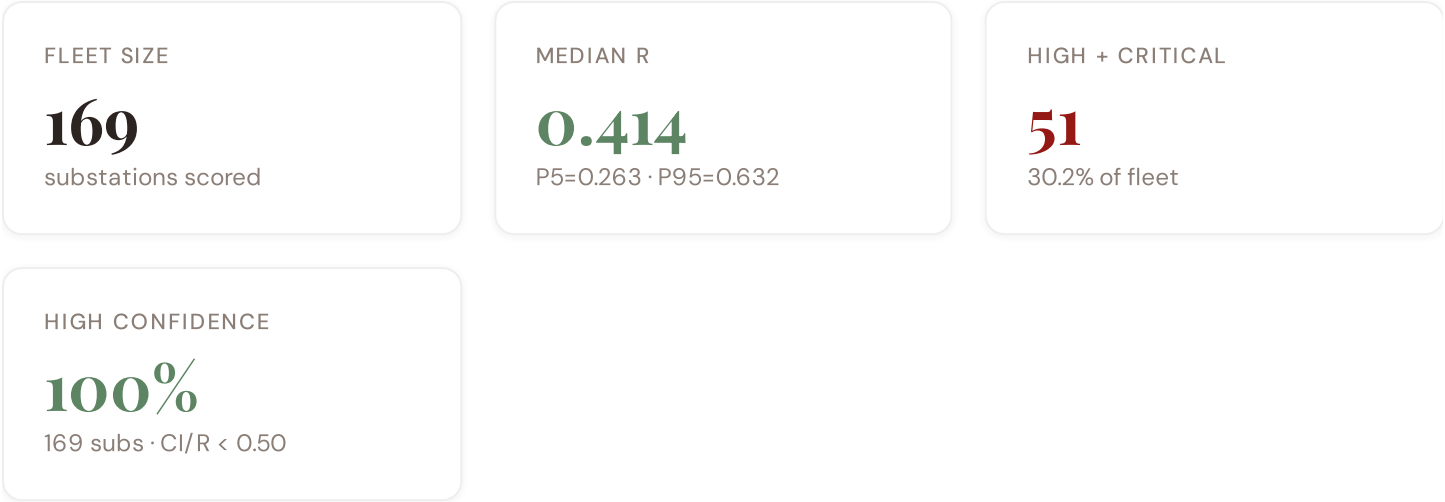
Data Source Registry — June 2026

d01_ice	ICE — Instituto Costarricense de Electricidad (vertically integrated TSO + dominant DSO + dominant generator)	DAILY	C2,C3,I1,I2,I3,R4,T1	16 vars
d06_imn	IMN — Instituto Meteorológico Nacional (under MINAE) + CNE Comisión Nacional de Emergencias	DAILY	I2,I3,R6_climate,R6_...	6 vars
d08_eor_epr	EOR + EPR — Ente Operador Regional + Empresa Propietaria de la Red (SIEPAC operations)	DAILY	I1,I2,S2	6 vars
d03_ovsicori	OVSICORI-UNA + RSN (UCR-ICE) — volcanic + seismic monitoring	CONTINUOUS	I1,I2,I3,R6_seismic,R...	12 vars
d05_osm	OSM Overpass — grid topology (ISO3166-1=CR area filter)	MONTHLY	I1,I2,I3,R4	8 vars
d07_copernicus	Copernicus ERA5 + CMIP6	MONTHLY	R6_climate	4 vars
d02b_dso	8 DSOs (ICE direct + CNFL subsidiary + ESPH + JASEC + 4 cooperatives Coopelesca/Coopeguanacaste/Coopealfaro Ruiz/Coopesantos)	QUARTERLY	C1,V1,V2,I2,I3	18 vars
d04_inec	INEC — Instituto Nacional de Estadística y Censos	QUARTERLY	E1,E2,S1,S2,S3	16 vars
d04b_bccr	BCCR — Banco Central de Costa Rica (CRC monetary authority)	QUARTERLY	E1,E2	5 vars
d02_aresep	ARESEP — Autoridad Reguladora de los Servicios Públicos (Ley 7593 / 1996)	ANNUAL	C1,C2,V1,V2,T1,E1	12 vars
d09_csirt_cr	CSIRT-CR — Centro de Respuesta de Incidentes de Seguridad Informática (MICITT Dirección de Gobernanza Digital)	ANNUAL	R7_cyber	5 vars
d10_iaea_oecd	IEA + OECD — energy benchmarks (Costa Rica 38th OECD member, accession 25 May 2021)	ANNUAL	T1,E1	6 vars
d11_minae_sepse	MINAE + SEPSE — Ministerio de Ambiente y Energía + Secretaría de Planificación del Subsector de Energía	ANNUAL	T1,R6_hydro_deficit	8 vars

C.1 Update Frequency Timeline



C.2 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **169 substation scores remain unchanged** this edition. The fleet median R of 0.414 (mean 0.425) reflects a distribution skewed toward the lower bands, with 1.2% in Low and 68.6% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

10 changes tracked across PO, P1, and P2 releases

CR-S33B-1	NEW	Post-IS-upstream-cleanup re-onboarding from clean Iceland template (commits c2e41e06 + e4c78f91)	KB §73
CR-S33B-2	NEW	CR fact card §1-§17 web-verified anchors wired across all 8 pages (ICE + ARESEP + 7 provincias + R6 modifiers)	KB §73
CR-S33B-3	NEW	R3 4-tier calibration — Industrial-High-Tech 1.05 (Heredia Intel/Boston Scientific) / Capital-Metro 1.04 (San José + Alajuela) / Commercial-Geothermal 1.03 (Cartago + Guanacaste) / Light-Rural-Coastal 1.02 (Puntarenas + Limón)	methodology

CR-S33B-4	NEW	R6_hurricane + R6_hydro_deficit NEW sub-patterns — first Caribbean tropical-cyclone + first El-Niño-anchored hydro-deficit modifiers in cohort (Otto 2016 + Nate 2017 + 2023-2024 El Niño anchors)	methodology
CR-S33B-5	ENHANCED	SIEPAC cross-border re-activation — CR-NI north + CR-PA south interconnects (first non-zero cross_border_lines in 3 onboardings)	methodology
CR-S33B-6	DATA	dO5_osm LIVE — ISO3166-1=CR area filter (SIEPAC + ICE backbone + Reventazón/Arenal/Miravalles/Las Pailas switchyards)	methodology
CR-S33B-7	DATA	ICE single-DSO + CNFL subsidiary + 2 municipals + 4 cooperatives + ARESEP regulator + OVSICORI-UNA/RSN + INEC/BCCR + IMN + EOR/EPR + CSIRT-CR/MICITT + MINAE/SEPSE wired	methodology
CR-S33B-8	ENHANCED	edition_anchor_month_offset=5 — cohort-synchronized Edition O2 = 2026-07-09 (BPG Discipline #20 PASS)	intelligence
CR-S33B-9	ENHANCED	D#21 content-leakage gate PASS (O IS + O HU hits; CR S33A baseline was 142 + 104)	KB §73
CR-S33B-10	NEW	R7 ceiling 1.025 — 2022 ransomware national emergency anchor (Conti Hacienda + Hive CCSS) + post-2022 hardening (CSIRT-CR + NCS 2023)	methodology

SECTION D — MONTHLY DEEP-DIVE

Cartago — the Pacific Ring of Fire Eruption Cycle & Miravalles

Deep-dive rotation: Each edition features a substantive thematic analysis. The 8-provincia rotation covers Costa Rica's most material narratives: **Ed. 01** Cartago — Pacific Ring of Fire eruption cycle 2021- + Miravalles 2023-2024 + Heredia industrial corridor evacuation · **02** San José — San José capital load + Miravalles/Las Pailas geothermal baseload · **03** Limón — Intel Heredia corridor + Reventazón 305.5 MW + Arenal hydro-deficit exposure · **04** Heredia — Coyoil Free Trade Zone (Alajuela) smelter + Sep 2024 IPP 166 MW solar+wind (target 2027) · **05** Alajuela — Arenal-Corobicí cascade hydro complex + Middle America Trench 2008 Mw 6.3 doublet + Cinchona 2009 2010 + Pirris hydro-deficit · **06** Guanacaste — Liberia + Miravalles/Þeistareykir geothermal + Costa Rica Deformed Belt seismic · **07** Limón — Westfjords peripheral grid + Caribbean hurricane (Otto 2016 + Nate 2017) exposure · **08** Puntarenas — Cachí 150 MW + rural radial network.

D.1 Why Cartago, Why Now

CORRIDOR SUBSTATIONS

12

12 EHV · 0 HV

HIGH BAND

6

50.0% of corridor

CORRIDOR MEDIAN R

0.536

Fleet median: 0.414

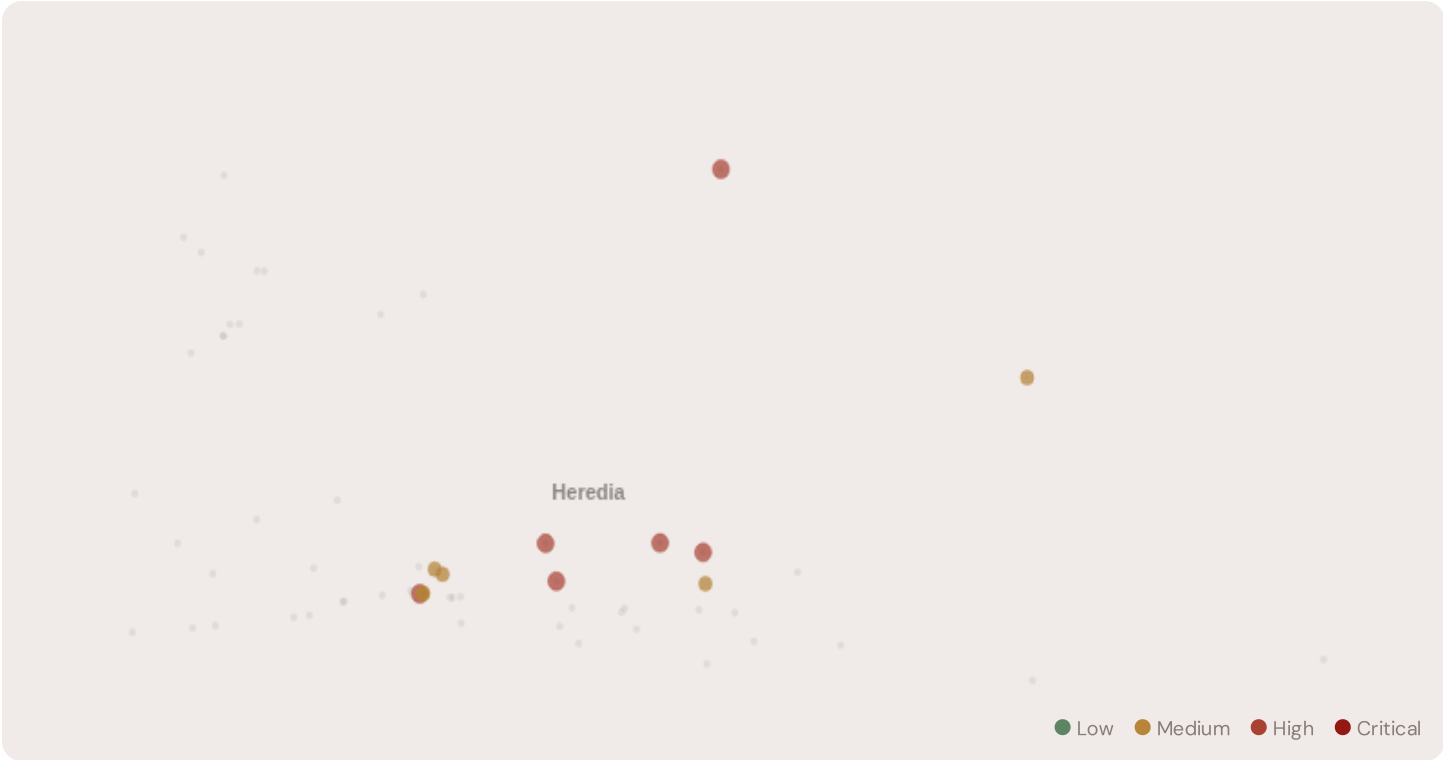
WORST PROVINCIA

Heredia

Avg R: 0.485 · 12 substations

The Heredia Tech Corridor — Intel + Boston Scientific (Edition 01 inaugural) hosts **12 substations** across 1 provincia regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **6 substations (50.0%)** are classified High, with only **0** in the Low band. 75% of corridor substations score above the national fleet median of 0.414. The corridor median R of 0.536 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCIA	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Substation w557557063	Heredia	0.628	High	0.417	0.621	0.255	0.628	0.553	0.088	
ICE – Central Telefónica	Heredia	0.586	High	0.462	0.614	0.278	0.586	0.516	0.098	
Subestación Chilamate	Heredia	0.581	High	0.393	0.671	0.254	0.581	0.526	0.083	
Subestación Heredia	Heredia	0.556	High	0.462	0.677	0.341	0.556	0.527	0.097	
CNFL	Heredia	0.537	High	0.460	0.664	0.327	0.537	0.594	0.099	
Subestación Concepción	Heredia	0.536	High	0.418	0.667	0.321	0.536	0.585	0.114	

SUBSTATION	PROVINCIA	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Subestación Belén	Heredia	0.497	Medium	0.442	0.621	0.329	0.497	0.544	0.108	
Substation w557557065	Heredia	0.427	Medium	0.407	0.658	0.284	0.427	0.542	0.101	
Subestación General	Heredia	0.415	Medium	0.425	0.631	0.288	0.415	0.589	0.097	
Subestación La Ribera	Heredia	0.370	Medium	0.374	0.683	0.273	0.370	0.593	0.111	

... 2 additional substations — [explore full corridor in Digital Twin →](#)

D.3 Component Analysis

COMPONENT AVERAGES — HEREDIA TECH CORRIDOR — INTEL + BOSTON SCIENTIFIC (EDITION 01 INAUGURAL) VS FLEET			MODIFIER AVERAGES — HEREDIA TECH CORRIDOR — INTEL + BOSTON SCIENTIFIC (EDITION 01 INAUGURAL) VS FLEET		
C — Continuity	0.419 vs 0.418 (0%)		R3 Consequence	1.050	fleet 1.035 ↑
I — Infrastructure	0.295 vs 0.298 (-1%)		R4 Graph Crit.	1.000	fleet 1.000 →
S — Saturation	0.558 vs 0.550 (+1%)		R6a Restoration	0.060	fleet 0.060 ↓
V — Voltage	0.648 vs 0.651 (-0%)		R6b Seismic	0.160	fleet 0.185 ↓
E — Economic	0.485 vs 0.425 (+14%)		R7 Digital Read.	1.025	fleet 1.025 ↑
T — Transition	0.097 vs 0.099 (-2%)				

The dominant risk driver across the Heredia Tech Corridor — Intel + Boston Scientific (Edition 01 inaugural) corridor is **V (Voltage)**, averaging 0.648 against a fleet average of 0.651 — occupying 648% of its weight ceiling (w = 0.10). The second contributor is **E (Economic)** at 0.485 vs fleet 0.425 (485% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.050 (fleet: 1.035), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 0.160, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 Provincia Breakdown

PROVINCIA RISK PROFILE — SORTED BY MEAN R	
Longer bar = lower R = better resilience.	
Heredia	12 subs · avg R = 0.485 · H:6 M:6

The provincia regions-level breakdown reveals significant intra-regional variation. **Heredia** leads with a mean R of 0.485 across 12 substations, while **Heredia** scores 0.485 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Heredia Tech Corridor — Intel + Boston Scientific (Edition 01 inaugural) is not uniformly

stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

O Heredia Tech Corridor — Intel + Boston Scientific (Edition 01 inaugural) substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Heredia sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

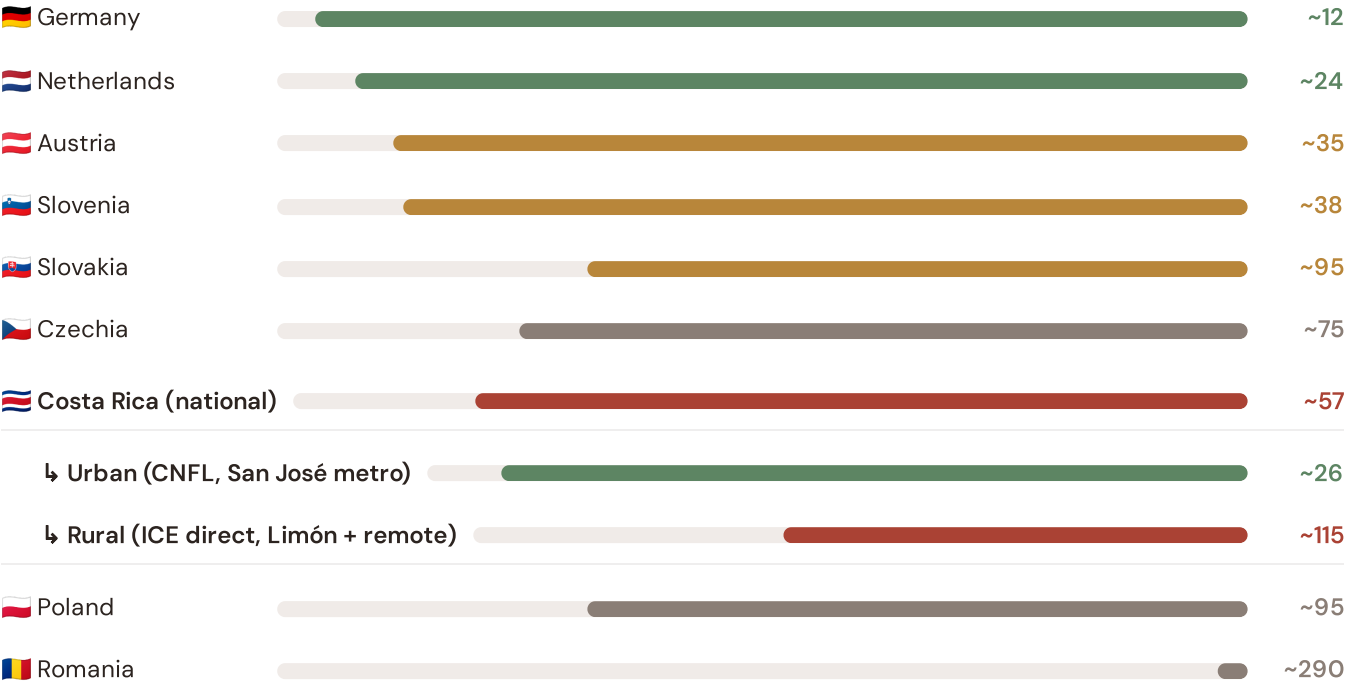
SECTION E — RECURRING MODULE

European Context Card

How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the —corridor analysis hinges on Costa Rica's continuity performance relative to European peers, especially the Visegrád-4 + CEE-South cohort siblings. Future editions rotate through: VoLL comparison (Ed. 02), DER saturation vs Germany (03), CMIP6 climate hazard vs DACH peers (04), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



Source: CEER 7th Benchmarking Report (2022); ARESEP (2024) quality reports; per-DSO filings (CNFL + ICE + Coopelesca + Coopeguanacaste 2024). Costa Rica sub-categories from ARESEP territory classification (San José metro under CNFL vs ICE direct rural Limón/remote). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

This month's key insight: The Heredia Tech Corridor — Intel + Boston Scientific (Edition 01 inaugural) corridor deep-dive shows **0 substations** (of 0) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.000 is **0.0× the fleet average** of 0.418. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 8 Costa Ricaic provincia

Where does Cartago sit within the national landscape? Provincia sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



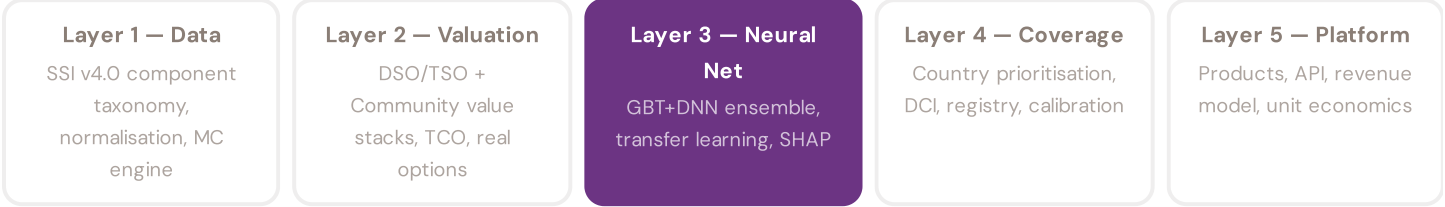


The Heredia Tech Corridor — Intel + Boston Scientific (Edition 01 inaugural) ranks **#0 of 7 provincia regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are heredia, cartago, guanacaste, while the three most resilient are san-jose, limon, puntarenas. 3 of 7 score above the fleet average of 0.425. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to CNFL, ICE direct and HS CNFL DSOs and to ICE — and to the community?"* Each edition explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3

§0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

- SHAP: game-theoretic feature attribution per prediction
- Waterfall: fleet average → substation prediction
- Direct traceability to SSI components and data sources
- Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 169 substations with complete feature vectors. Average CI_width of 0.110 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 2 Low, 116 Medium, 51 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 3** Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Deep-Dive: Guanacaste

Deep-dive into Guanacaste's grid risk profile — substation map, component analysis, provincia regions breakdown.

NEXT DATA REFRESH

Q3 2026 — 3 Quarterly Sources

DSO filings (CNFL + ICE direct + ESPH + JASEC + 4 cooperatives), INEC, BCCR are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **ARESEP — Autoridad Reguladora de los Servicios Públicos (Ley 7593 / 1996)** (12 variables → C1,C2,V1,V2,T1,E1).

FLEET SPOTLIGHT

Heredia: Highest Risk Concentration

Heredia has the highest concentration of High/Critical substations: **6 of 12 (50%)** are in the upper risk bands — avg R = 0.485. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Guanacaste**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent an Costa Ricaic utility, ARESEP, ICE, ICE generation fleet, ICE geothermal, Las Pailas IPP, CNFL, ICE direct, HS CNFL, IMN, MICITT (ECOI), CSIRT-CR, or an academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Costa Rica Edition 01 (Inaugural)

Published 13 August 2026 · SSI Index v4.0.2 · 169 substations · ~1,428 grid lines · 6 components · 20 metrics · 7 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5

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